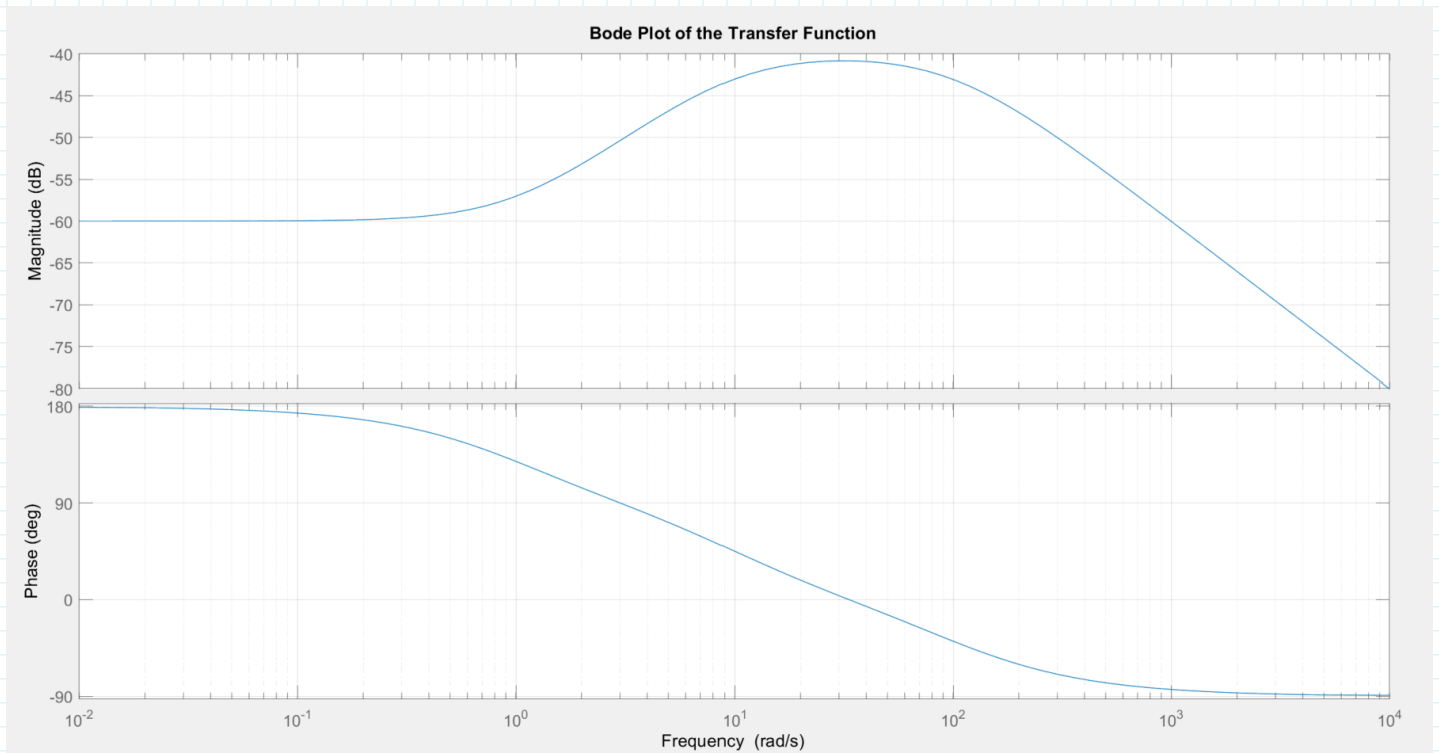
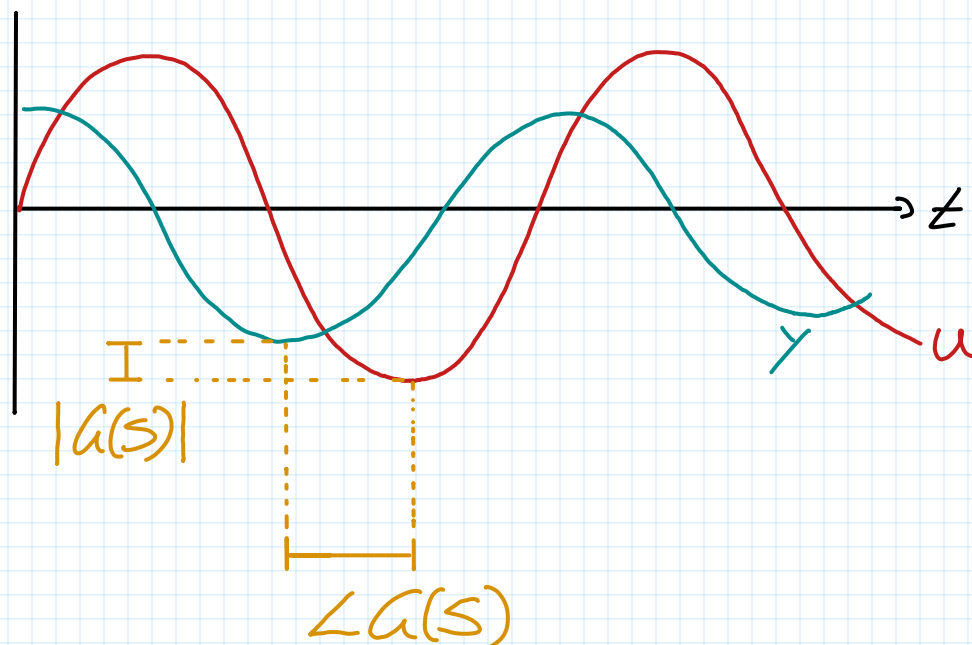


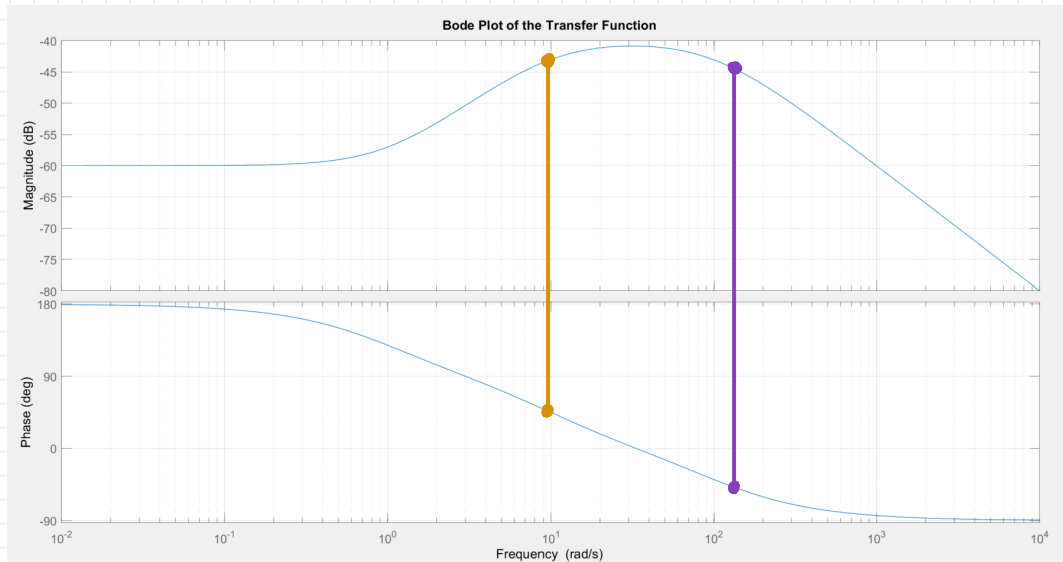
• Last time we looked at the Bode plot.



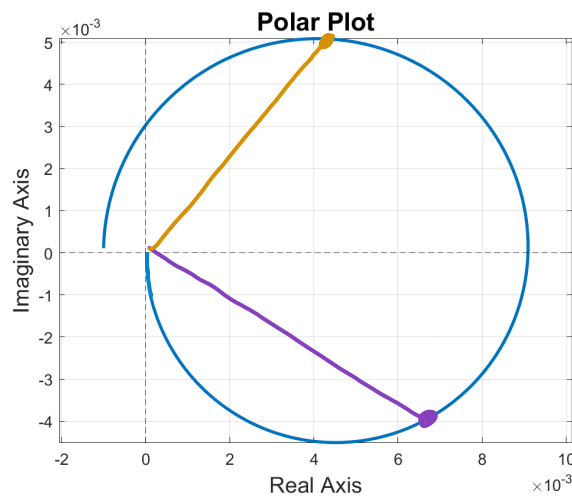
A set of graphs that show magnitude and phase of a system, depending on the frequency input.



Then we saw, that we can summarize both plots in one, by making the frequency



Make ω implicit



This is where we left off. The Polar plot can now be extended to become really useful for stability analysis!

Why do we need something other than Bode plots?

+ Intuitive display of frequency response

- Misleading if

•

•

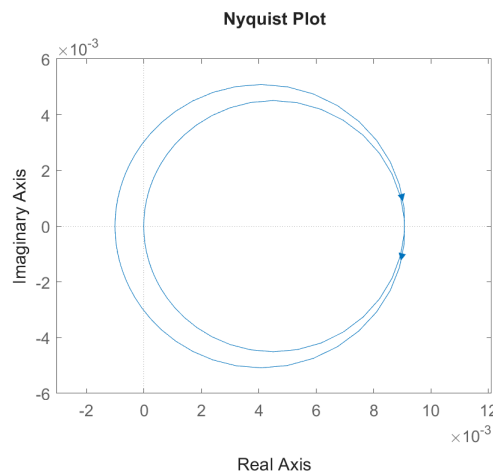
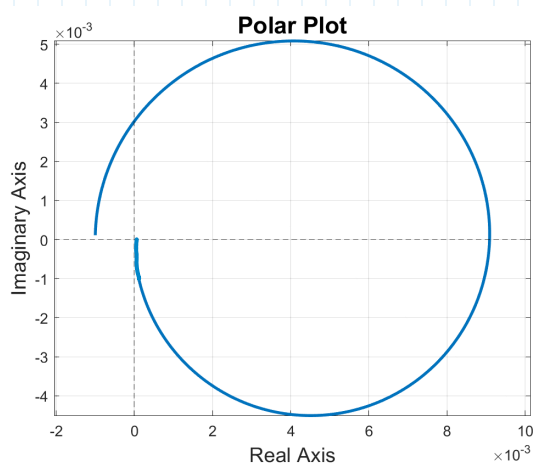
} Nyquist plot is
⇒ Solves these!

Nyquist Plot

• The polar plot goes from ∞ to 0.

⇒ The Nyquist plot extends this to $\omega \in \mathbb{R}$.

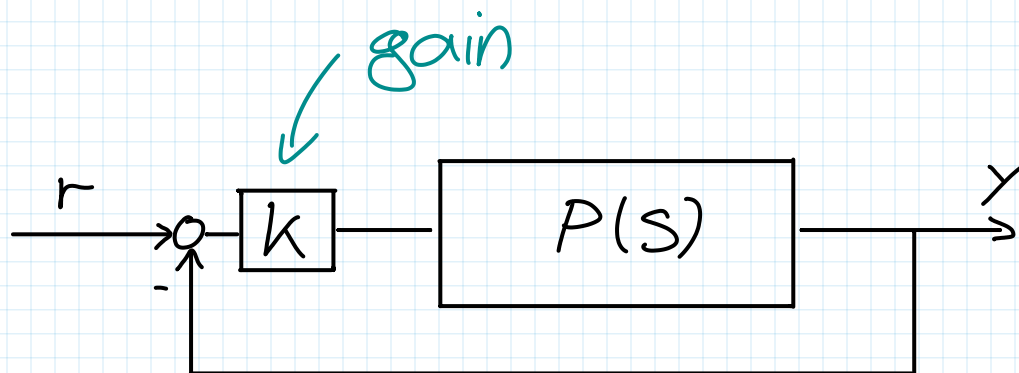
⇒ Graphically this is equivalent to mirroring the polar plot about the real axis.



⇒ Not a lot of new information right?

The Nyquist plot will help us assess closed-loop stability based on the open-loop

Recall standard feedback architecture:



The closed loop has tf $T(s)$

$$T(s) = \frac{kP(s)}{1 + kP(s)}$$

The closed loop is **unstable**, if $T(s) = \infty$.

$$T(s) = \infty \text{ if}$$

$\Rightarrow$ System blows up, if $P(s) =$

The $-\frac{1}{k}$ point will be very important!

Note how we are making conclusions about closed-loop stability, based on the open loop!

To determine stability of a closed loop system, we look at the:

Given a Nyquist contour Γ_s , let P be the number of poles of $G(s)$ encircled by Γ_s , and Z be the number of zeros of $1 + G(s)$ encircled by Γ_s . Alternatively, and more importantly, if Z is the number of poles of the closed loop system in the right half plane, and P is the number of poles of the open-loop transfer function $G(s)$ in the right half plane, the resultant contour in the $G(s)$ -plane, $\Gamma_{G(s)}$ shall encircle (clockwise) the point $(-1 + j0)$ N times such that $N = Z - P$.

~ Wikipedia

• Largest point of confusion in CSI...

Let's break it down:

• This equation always holds:

Z : # of

P : # of

N : # of

of the point.

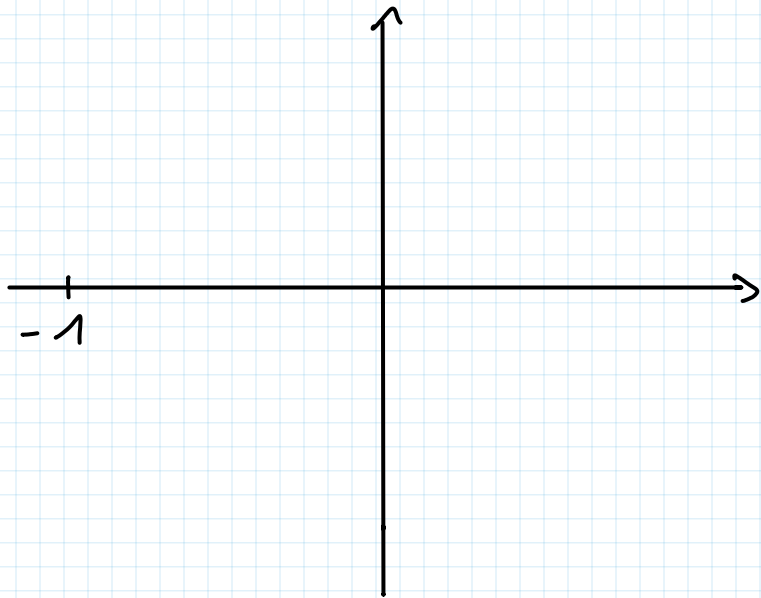
⇒ For a stable closed loop, we require

• Therefore we need

Ex:

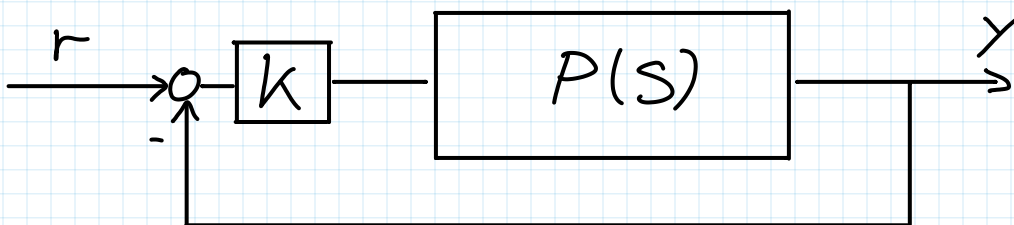
Given $P(s) = \frac{1}{s-2}$ } $\Rightarrow$ unstable pole!

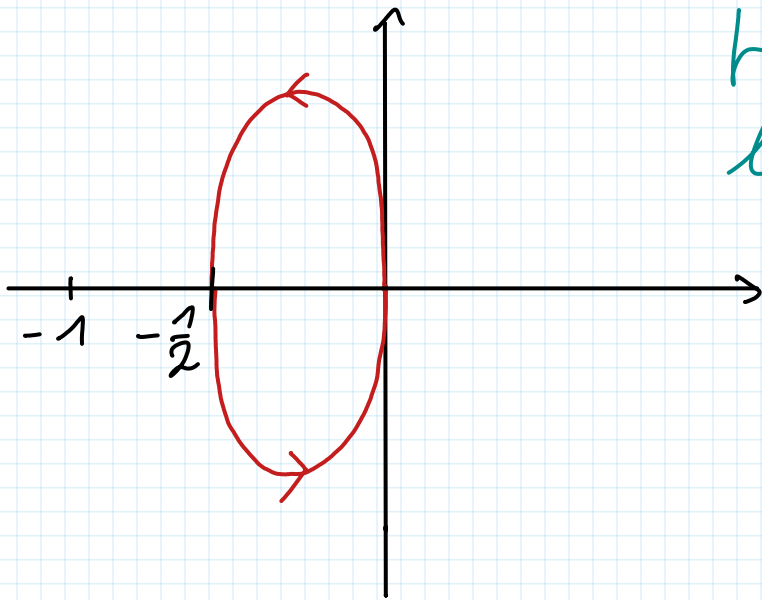
Plotting Nyquist of $P(s)$ we get:



For stability we require

We need *Nyquist* to encircle $-\frac{1}{k}$ -1 times (or once in counter-clockwise direction!)





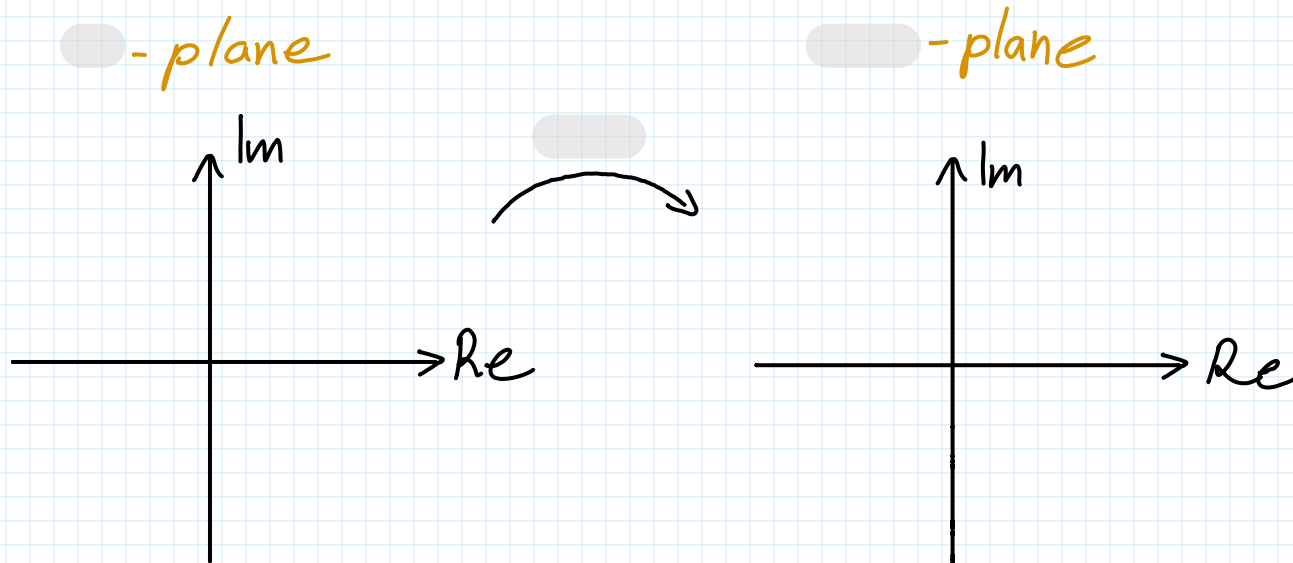
Here: $-1/u$ must lie in **this** realm

Condition for stable closed loop:

•
•
 $\Rightarrow$ Unstable open loop is stabilized when in closed loop configuration for .

Why this works:

• $G(s)$ "maps" points:



• The mapping is characterized by $|G(s)|$ & $\angle G(s)$

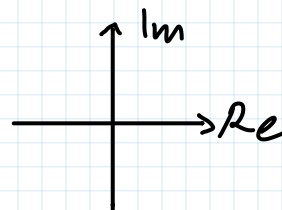
$$\Rightarrow |G(s)| =$$

$$\Rightarrow \angle G(s) =$$

So for the mapping of a point s , we must evaluate its phasors with respect to each pole / zero!

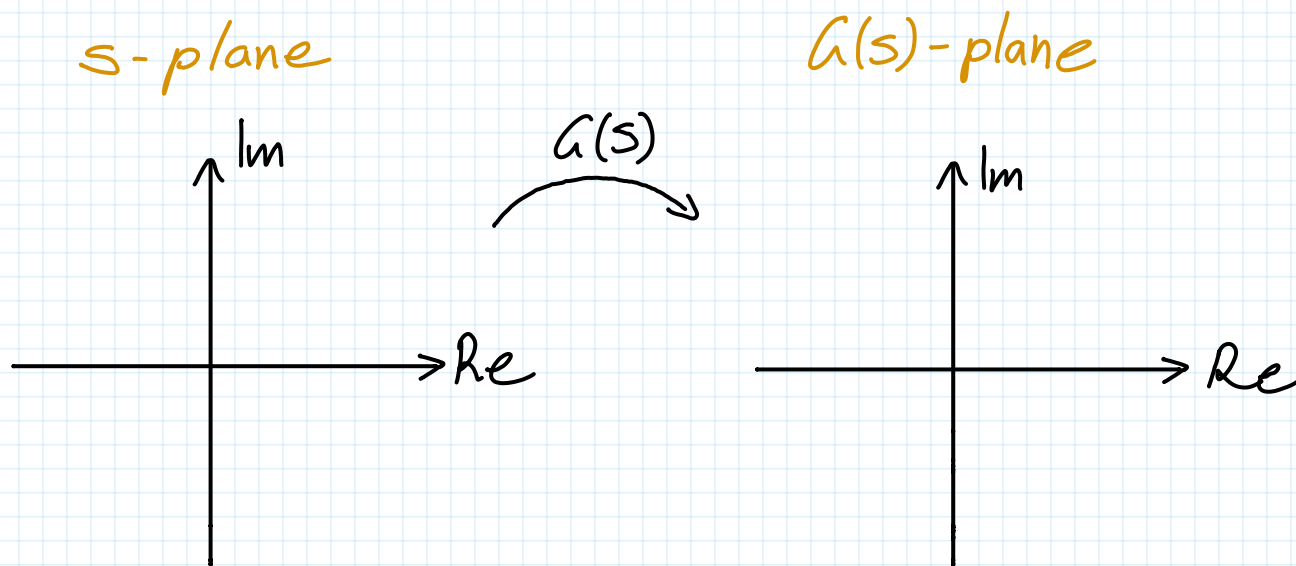
Example:

$$G(s) = \frac{s+2}{s+1}$$



$$\bullet |G(s)| = \frac{|s+2|}{|s+1|} \quad , \quad \bullet \angle G(s) = \angle(s+2) - \angle(s+1)$$

Now let's see how points are mapped:



$$|G(-1+j)| =$$

$$\angle G(-1+j) =$$

$$=$$

Core takeaway, zeros "add" phase, poles "subtract" it.

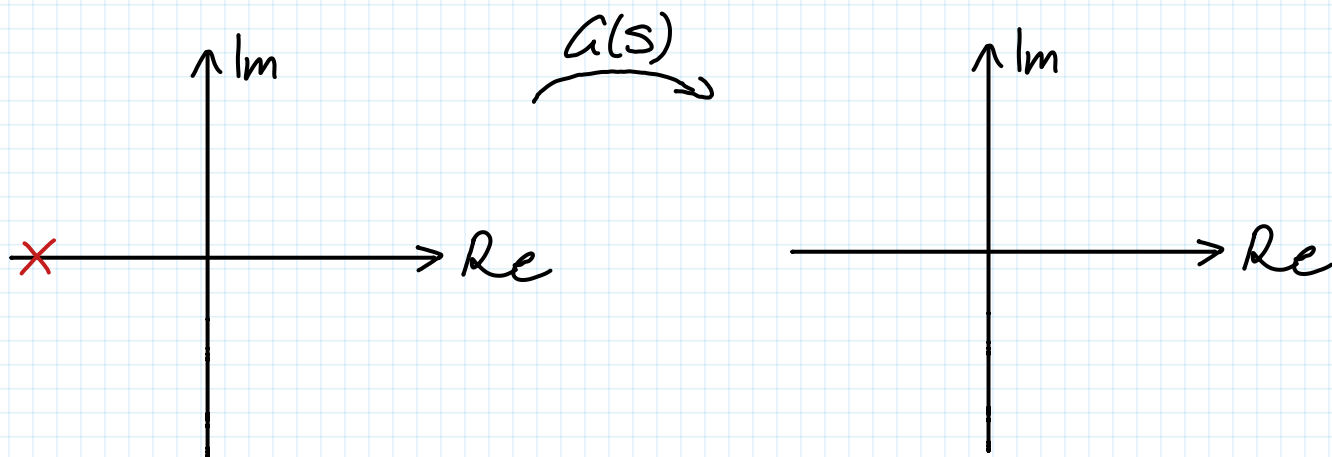
- We can now do this for many points. Especially interesting results will arise when we do this for infinitesimally close points, along a curve.

$\Rightarrow$

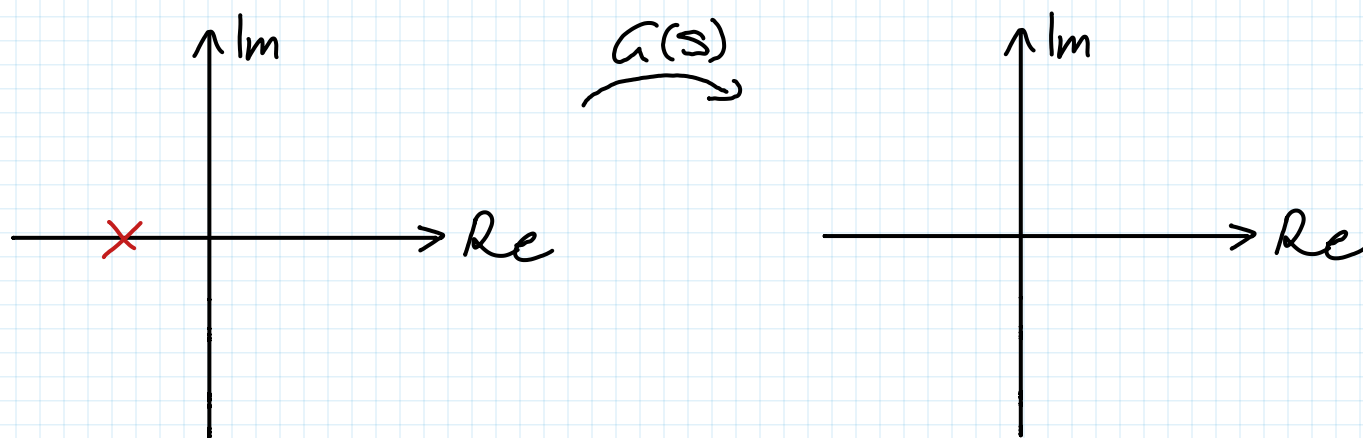
Let's start with a simpler system with a single pole p .

$$G(s) = \frac{1}{s-p}$$

First, let's draw a contour Γ that doesn't encircle the pole.

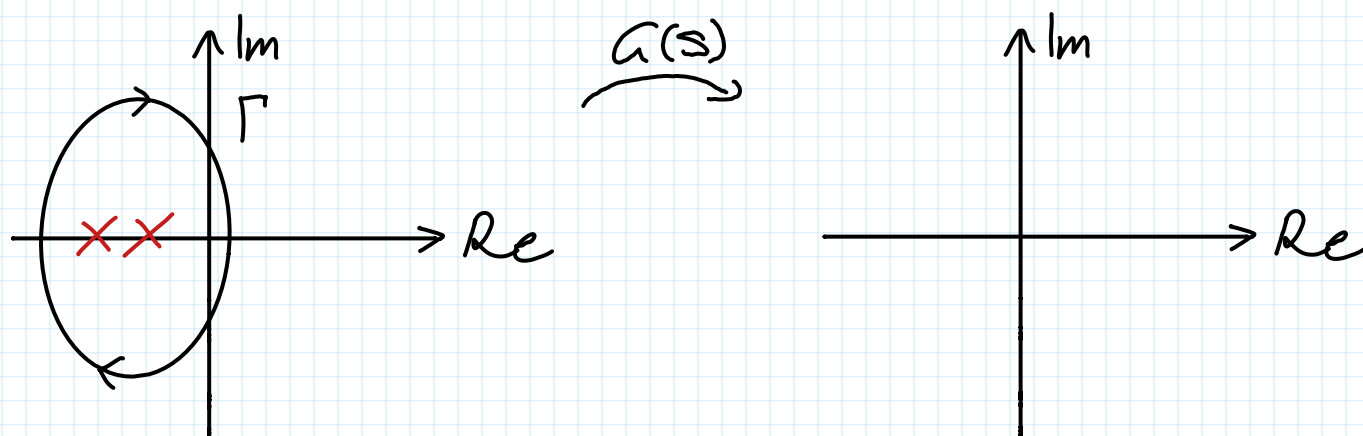


• If we choose the contour Γ to encircle the pole...



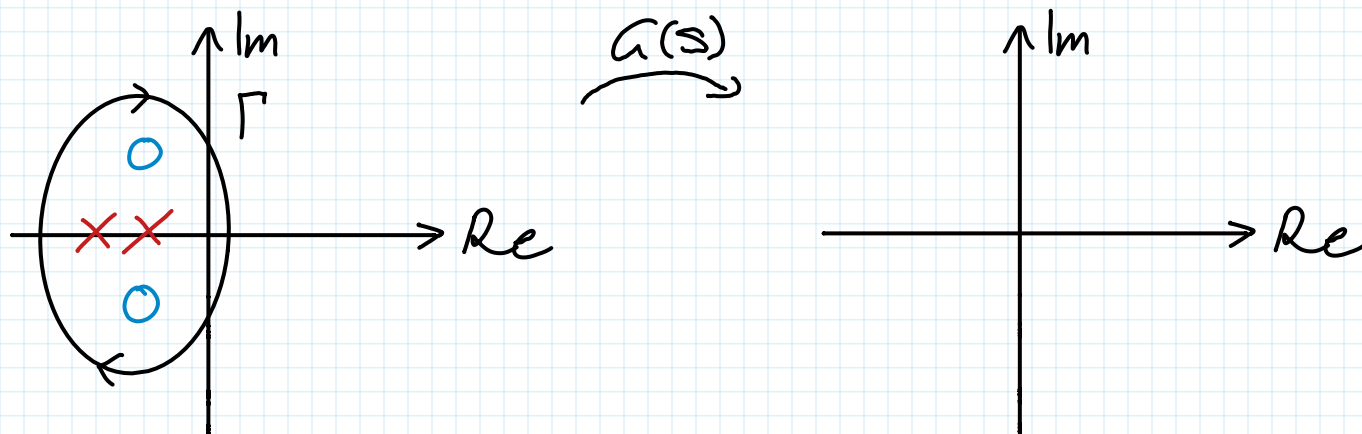
Γ' will encircle the origin in the opposite direction!

• What if we add two poles?



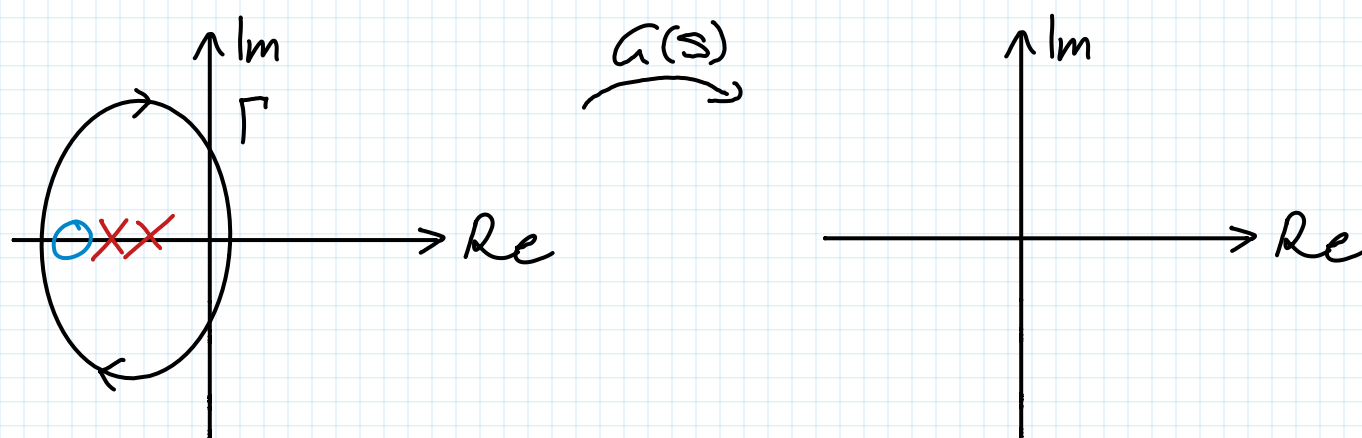
⇒ Two encirclements of the origin!

• And what if there are zeros?



Here the zeros add and the poles subtract. Therefore there are

• So what if we have an unequal amount of p_i and z_i ?



We conclude:

$$N = Z - P$$

Number of times $G(s)$ encircles origin $\rightarrow N$

Number of zeros of $G(s)$ $\rightarrow Z$

Number of poles of $G(s)$ $\rightarrow P$

Okay, this is neat - but how does it help us design controllers?

From Cauchy's Argument principle to Nyquist's stability theorem

- Recall the closed-loop tf

$$T(s) = \frac{kL(s)}{1 + kL(s)}$$

Note: Here for simplicity
 $L(s) \hat{=}$ Plant, dynamic part
 of controller

Unstable if

But poles of $T(s)$ are really the

$$\begin{aligned} &\# \text{RHP zeros of } 1 + kL(s) \\ &\quad \hat{=} \\ &\# \text{unstable closed-loop poles} \end{aligned}$$

Core idea:

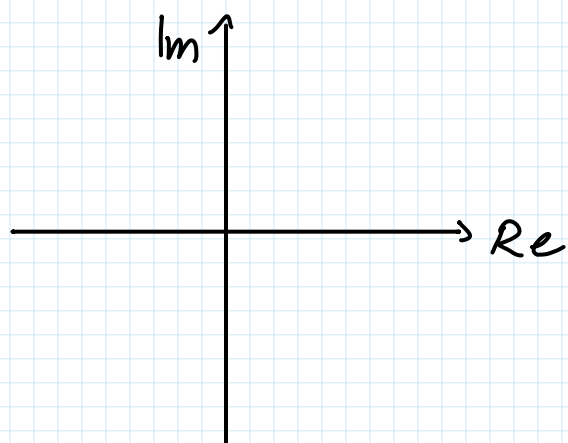
Use Cauchy's Argument principle, to check for RHP zeros of $1 + L(s)$:

$\Rightarrow$ Draw a contour in the complex plane, that entails information about the total RHP...

=> Solution: Choose the -contour or -contour:

s-Plane

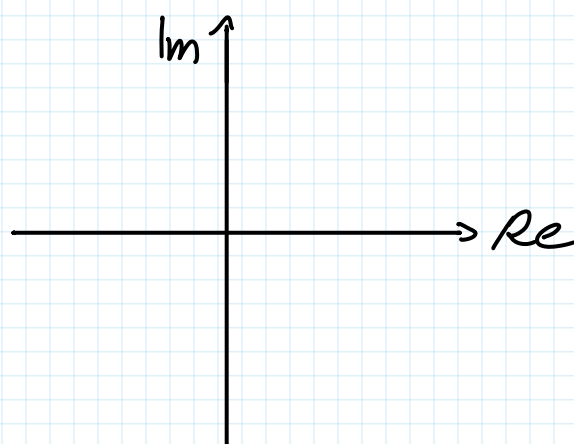
Close the contour at



=> Now choose the mapping as

$$G(s) = 1 + kL(s)$$

G(s)-plane



From $N = Z - P$:

Encirclements of of $1 + kL(s)$ plot

$\hat{=}$
of $1 + kL(s)$ - of $1 + kL(s)$

• This can be reformulated to become more practical.
First, notice that looking for solutions of

$\Leftrightarrow$

Therefore we can rewrite Simplifies (I)

$$\begin{aligned} & \# \text{Encirclements of origin of } 1+kL(s) \text{ plot} \\ & \quad \hat{=} \\ & \# \text{Encirclements of } \text{ } \text{ point of } \text{ } \text{ plot} \end{aligned}$$

• Secondly, recall that we can write

$$L(s) =$$

And therefore

$$1+kL(s) =$$

$\Rightarrow$ Denominator (poles) of $L(s)$ are
by closing the loop

Therefore we can write:

$$\begin{aligned} & \# \text{RHP poles of } 1+kL(s) \\ & \quad \hat{=} \\ & \# \text{Unstable } \text{ } \text{ poles} \end{aligned} \quad \text{Simplifies (II)}$$

• Finally, from the closed-loop tf

$$T(s) = \frac{kL(s)}{1+kL(s)}$$

We have that _____ of $1 + kL(s)$ correspond to the _____ of $T(s) = \frac{kL(s)}{1 + kL(s)}$.

RHP zeros of $1 + kL(s)$

$\hat{=}$

Unstable _____

Simplifies (III)

Let us combine everything to a final statement:

Nyquist stability criterion

N : # of encirclements of $-\frac{1}{k}$

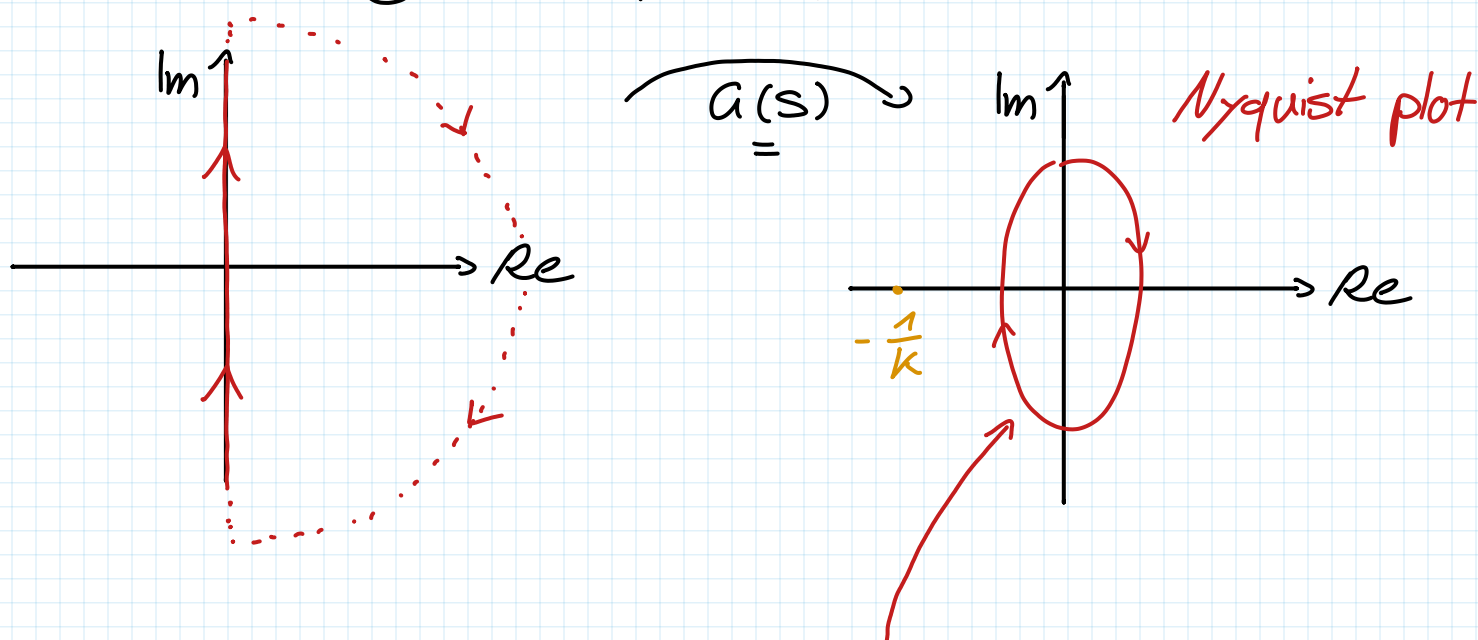
P : # of unstable open-loop poles

Z : # of unstable closed-loop poles

$\Rightarrow$ _____ closed-loop stability can be assessed using the _____ open-loop system!

That was a lot. Let's summarize the important points.

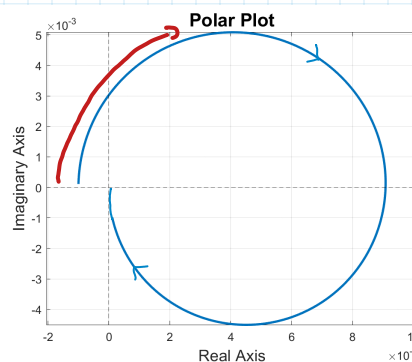
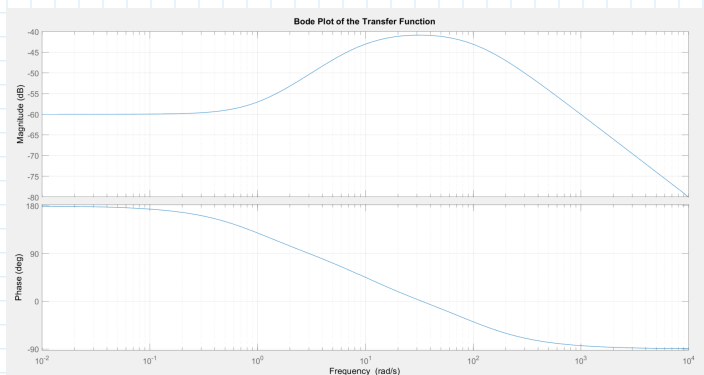
- When choosing to map a Nyquist-contour via $G(s)$

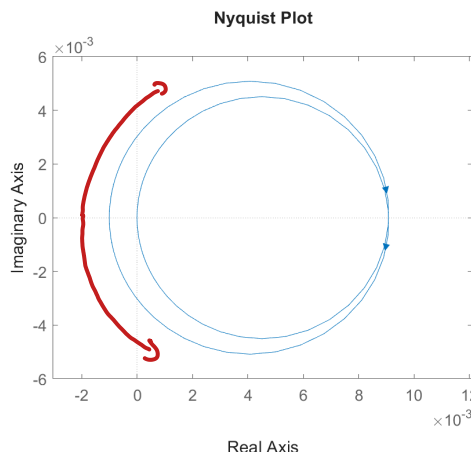
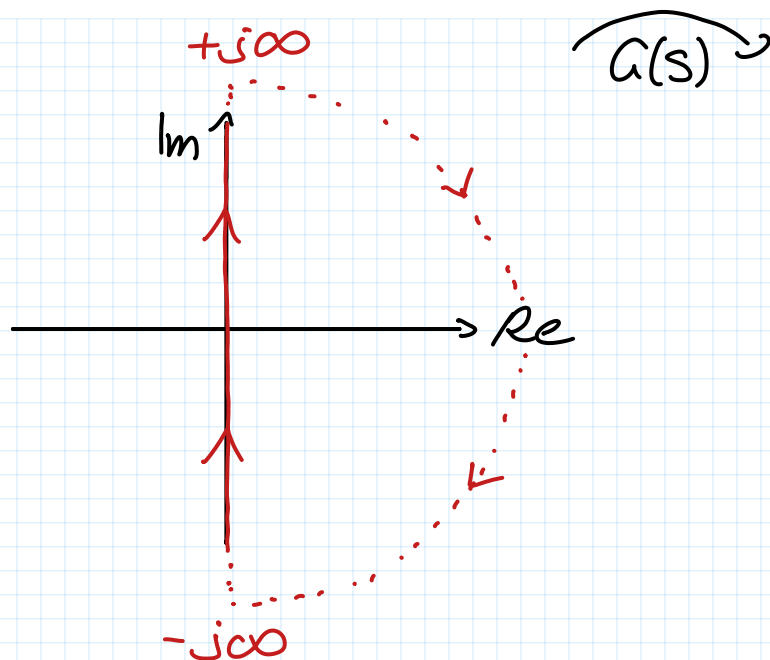


we get a shape called the Nyquist plot, that depends on the poles and zeros of $G(s)$.

- Now we can assess stability of the closed-loop, by evaluating the Nyquist criterion $N = Z - P$.

Recall: Earlier we said that the Nyquist plot is just an extension of the polar plot, which itself is just a reformulation of the Bode plot...





$\Rightarrow$ Mirroring the polar plot to get the Nyquist plot is exactly the same as mapping the Nyquist-contour!

In essence

- We want to assess the closed-loop, based on the open-loop frequency response.

$\Rightarrow$ Bode plot works often, but not always!

- So we introduced the Nyquist plot.
- Using the Nyquist stability criterion on the Nyquist plot, closed loop stability can always be assessed!

Why not just use root locus?

- + Nyquist also works when only present.
- + Root locus can't give insights on .
- + Nyquist can even handle transfer functions.

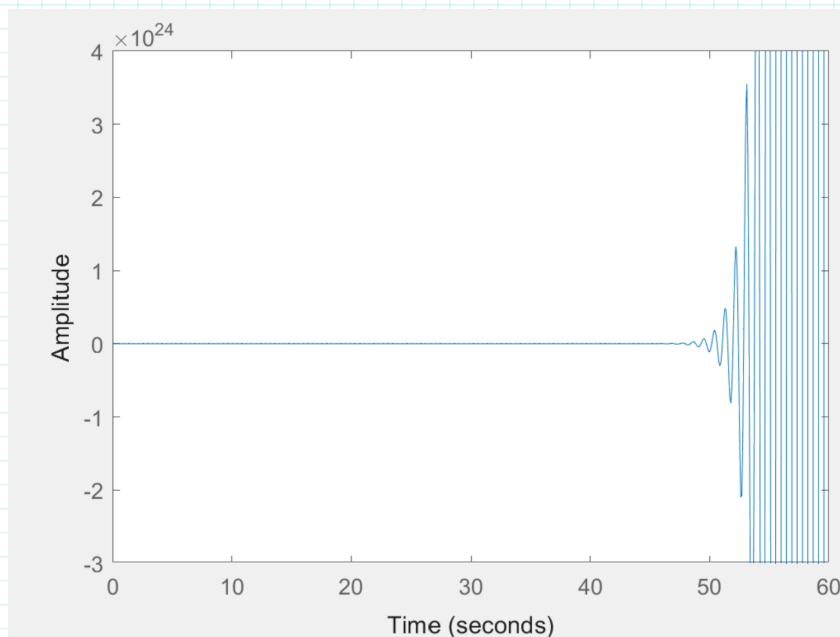
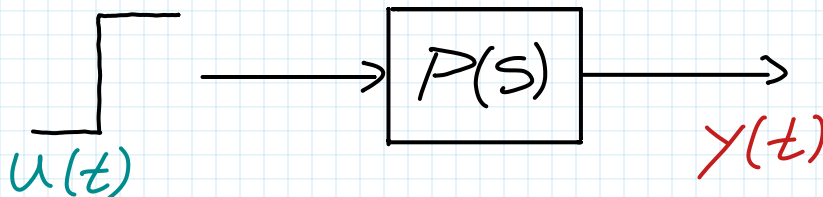
Another example:

$$\text{Given } P(s) = \frac{s^2 + 2s - 3}{s^3 - 2s^2 + 50s + 10}$$

zeros: $z_1 =$, $z_2 =$

poles: $p_1 \approx$, $p_2 \approx$, $p_3 \approx$

- Unstable poles, non-minimum phase zeros
 $\Rightarrow$ overall undesirable behaviour



Step response
 $y(t)$ of $P(s)$

• Can we stabilize this?

$\Rightarrow$ Apply Nyquist: $N = Z - P$

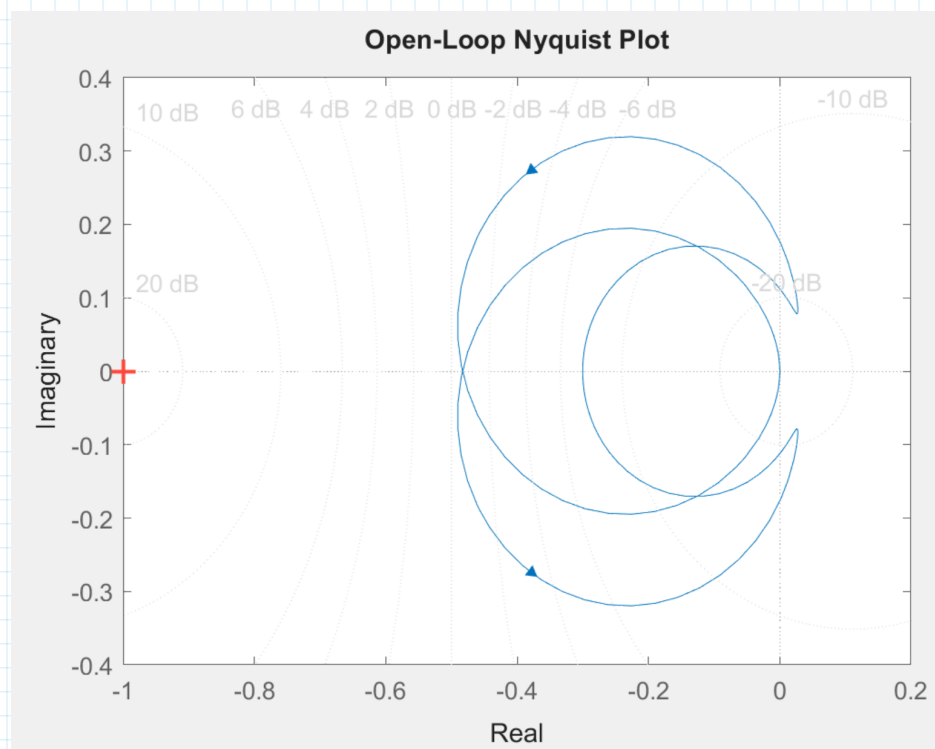
For closed-loop stability we need:

We have 2 unstable poles:

Therefore we need:

(2, counter-clockwise)

• Now we examine the Nyquist plot of $P(s)$:



$\Rightarrow$ To stabilize, $-1/k$ must be within this area

From the plot we approximate

$$\bullet -\frac{1}{k} < \quad \Rightarrow$$

$$\bullet -\frac{1}{k} > \quad \Rightarrow$$

Matlab

$\Rightarrow$ System is stabilized with a simple scalar gain $k \in [\text{ }]$ in a standard feedback configuration

With a perfectly known system and no disturbances this works great.

In reality though, we must not only ensure stability, but also robustness.

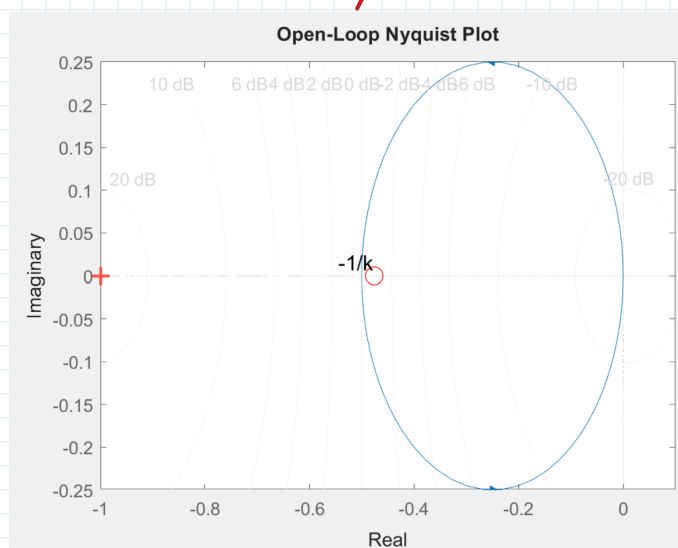
Robustness

This term refers to "how" stable our system is. A system is robust, if large and don't make it unstable.

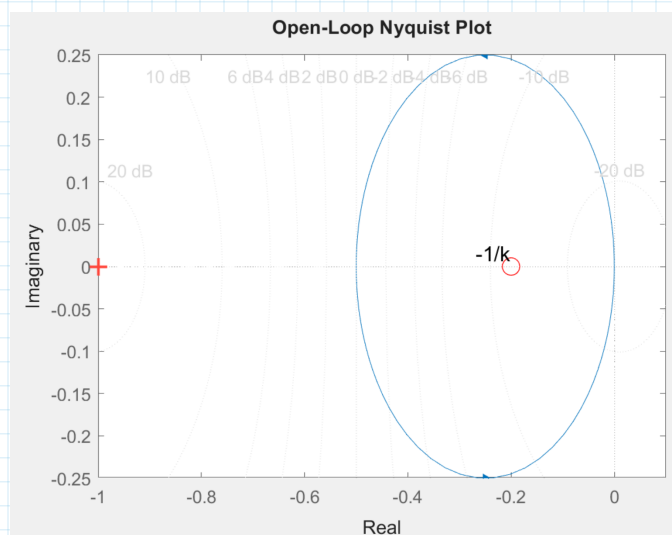
- Robustness can be measured by the Nyquist plot comes to the .

Ex: $P(s) = \frac{1}{s-2}$

$k = 2,05$



$k = 5$

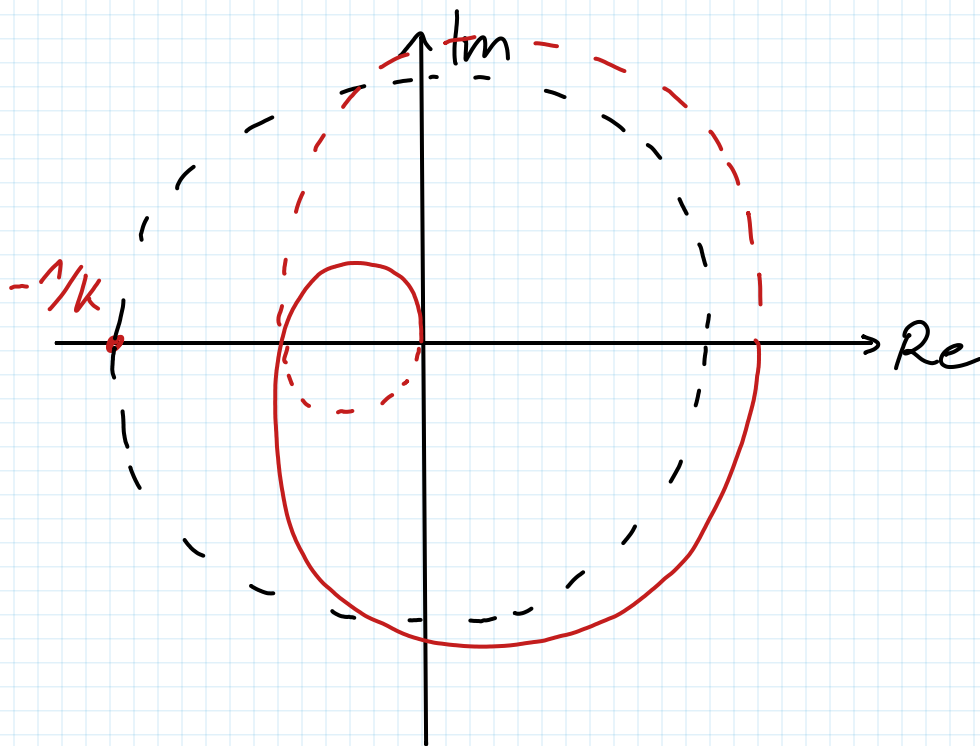


system becomes unstable for $k =$, but not for $k =$

• This "degree of stability" can be measured in the form of so called

Let's visualize these margins with an example:

Assume an open-loop stable system
 $\Rightarrow P = 0$, hence to remain stable
 $(Z = 0)$, we need $N = 0$.



If the plot encircles $-1/k$ the closed loop becomes unstable!

$\Rightarrow$ We measure the "safety distance" to $-1/k$ with:

g_m :

φ_m :

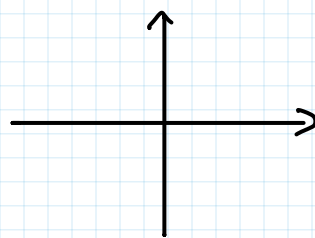
We want to keep the margins !

- Gain and phase margins can also be observed in the open-loop Bode plot.

⇒ The system operates at the unstable -1 point, if:

- $|L(s)| =$

- $\angle L(s) =$



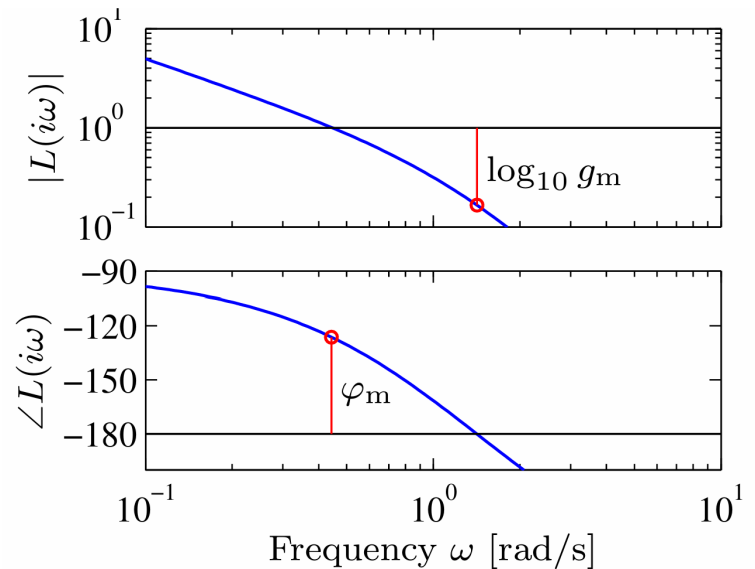
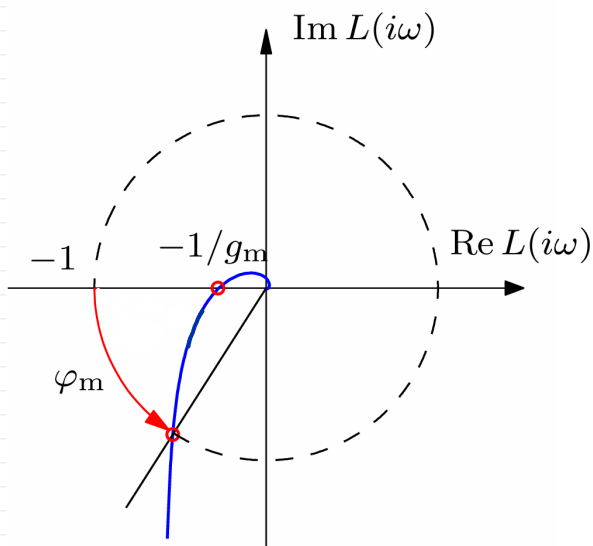
- So to ensure stability, we need

$$|L(s)| < \text{ }, \text{ whenever } \angle L(s) = \text{ }$$

- On the Bode plot, we see the margins as such:



Bringing it all together:



Now you know all the major plots in CSI:

- Root locus
- Bode
- Nyquist

Together they hold great power to analyze systems.

Me
explaining
Nyquist

